Extracting Trial-Varying Latent Variables in Cognitive Models with Neural Estimators

Ti-Fen Pan

UC Berkeley

SBI Summer school, July 2026

Lecture + Tutorial

01

What use cases can the tool (LaseNet) be applied?

Motivation, use cases, and workflow

02

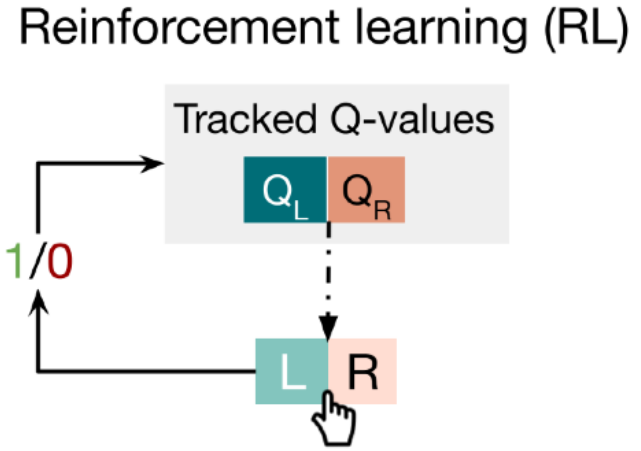
Hands-on tutorial

Try out a toy example in a notebook.

Trial-varying Continuous Latent Variables

Use Case 1 (continuous variable) :

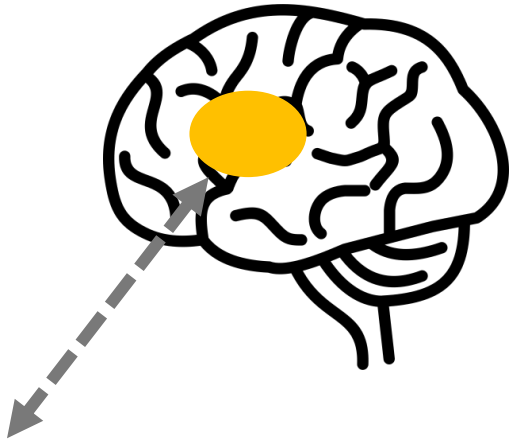
Reward prediction error in a Q-learning model



$$\delta_t = r_t - Q_t(a_t)$$

$$Q_{t+1} = Q_t(a_t) + \alpha * \delta_t$$

$$P(L) = 1/e^{\beta(Q(R)-Q(L))}$$



δ_t: reward prediction error at trial t (trial-varying latent variable)

r_t : reward at time t (observable variable)

a_t: action at time t (observable variable)

α: learning rate (model parameter)

β: choice inverse temperature (model parameter)

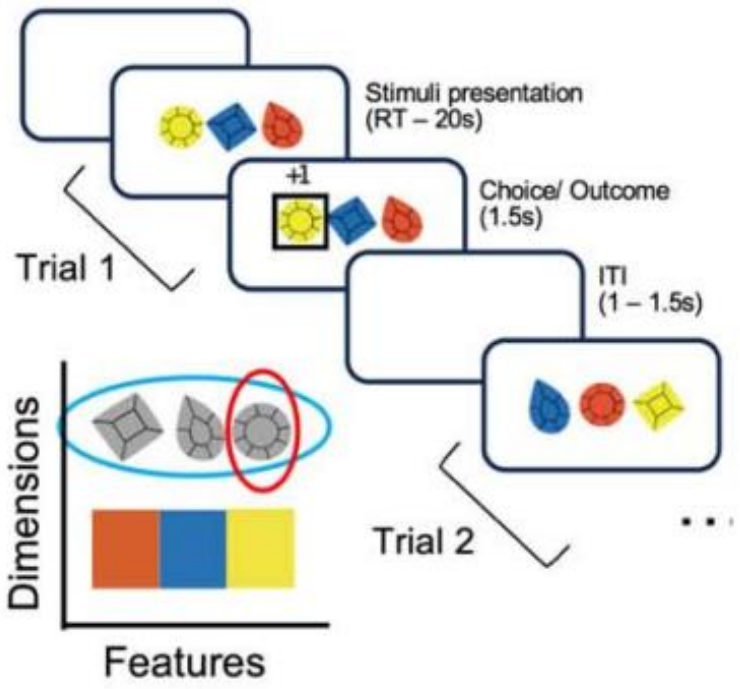
Schultz et al (1997)

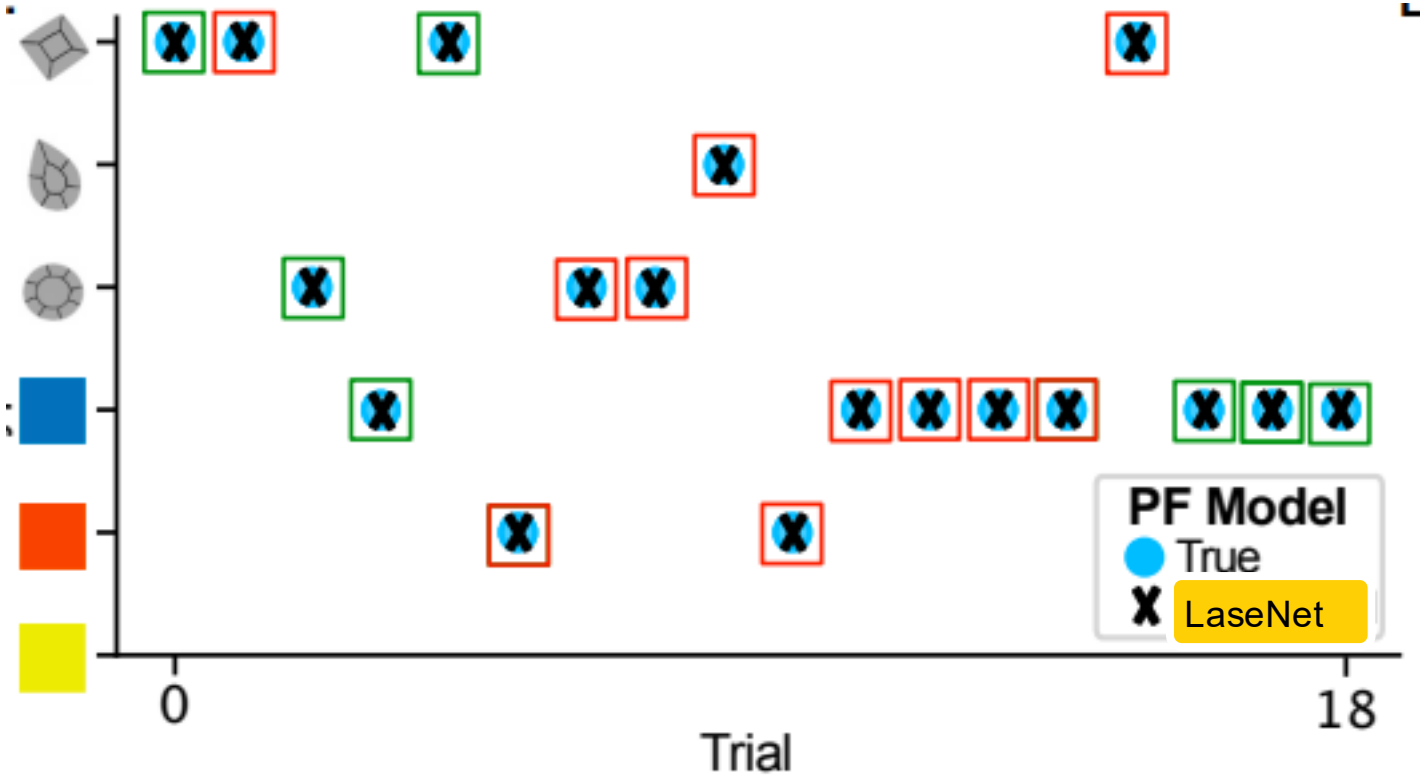
Gershman et al (2024)

Trial-varying Discrete Latent Variables

Use Case 2 (discrete variable) :

Latent stimuli feature - trial-by-trial attentional dynamics





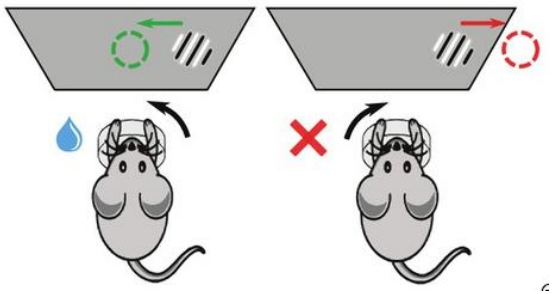
Maher et al (2025)

Learning to Attend Through Value-Based Hypothesis Testing

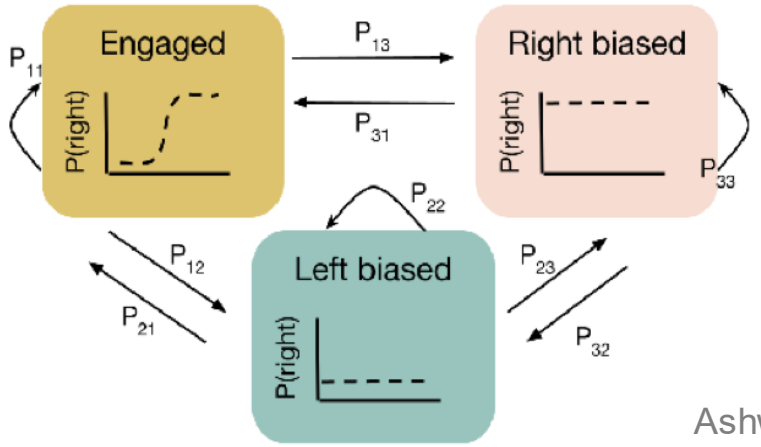
Trial-varying Discrete Latent Variables

Use Case 3 (discrete variable)

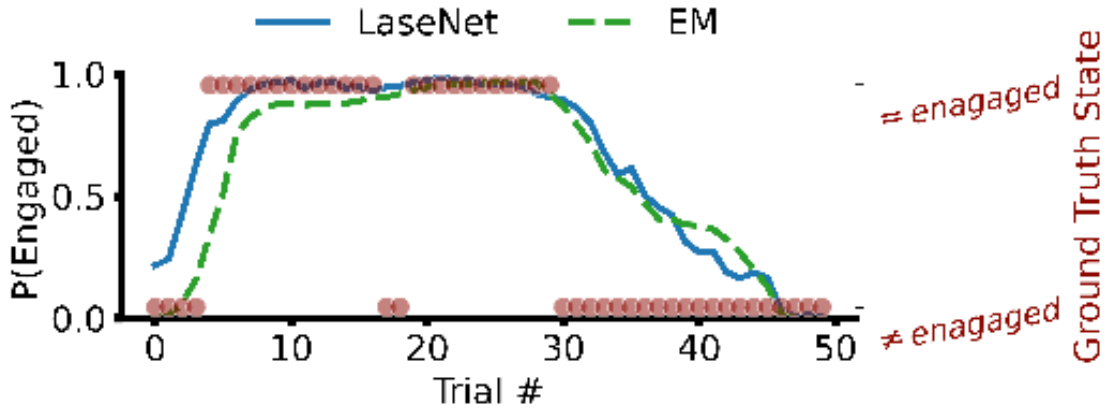
Latent states in a Bernoulli generalized linear model + Hidden Markov Model (GLM-HMM)



The International Brain Laboratory et. al (2021) eLife



Ashwood et.al. 2022
Pan et. al 2025



Standard Two-Steps Fitting Method

Maximum Likelihood Estimation (MLE)

01 Finding the best-fitting model parameters θ.

$\theta_{MLE} = \underset{\theta}{\operatorname{argmax}} P(Y|\theta)$,

$Y = (y_1, y_2, \ldots, y_T)$, y_t includes a set of stimuli, actions, and rewards (observable variables)

$P(\theta|Y) \propto P(Y|\theta) * P(\theta)$

Posterior $\propto$ Likelihood * Prior

02 Infer the latent variables z_t by running the computational model over participants' experienced sequences of Y with the fitted θ.

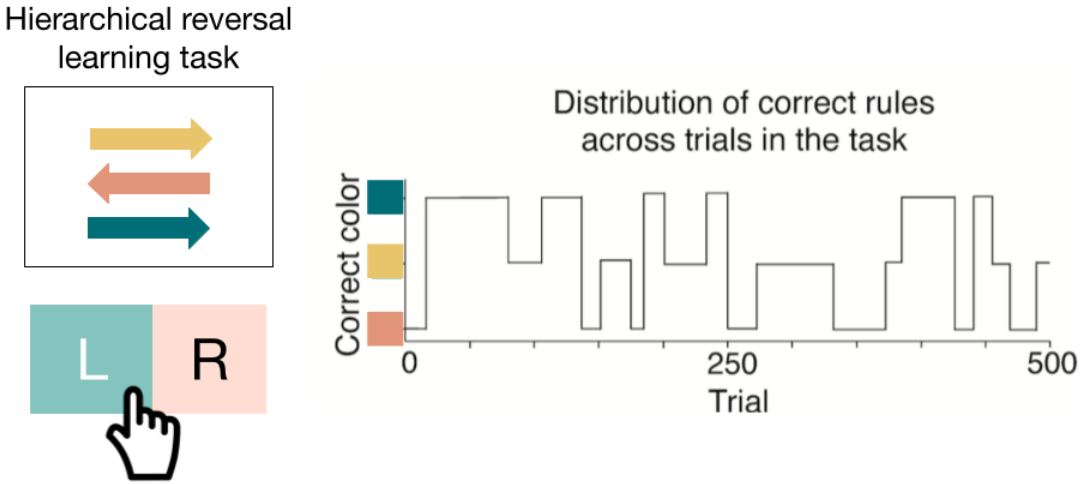
$z_t = f(z_{t-1}, \bar{y}_{t-1}, \theta_{MLE})$,

$\bar{y}_{t-1}$: the history of Y up to trial t-1

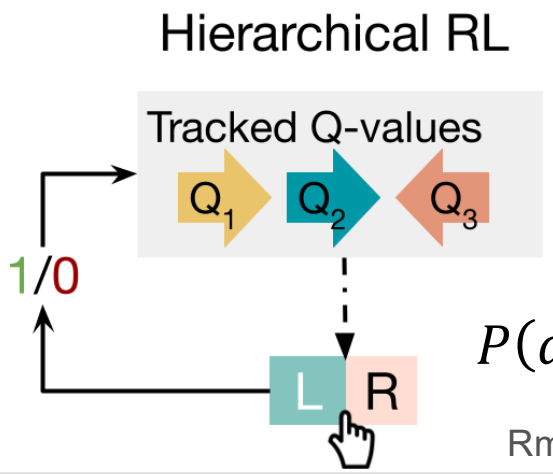
Target Joint Density: $P(z|\theta, Y)\ P(\theta|Y)$

Intractable likelihood

Hierarchical Reinforcement Learning (HRL)



Trial	Actions	Rewards	Chosen arrow seq
1	L	0	Red
…	…	…	…



$$P(arrow_i) \propto \exp(\beta * Q_t\ (arrow_i))$$

Rmus et.al 2024

What do researchers do if having intractable likelihood?

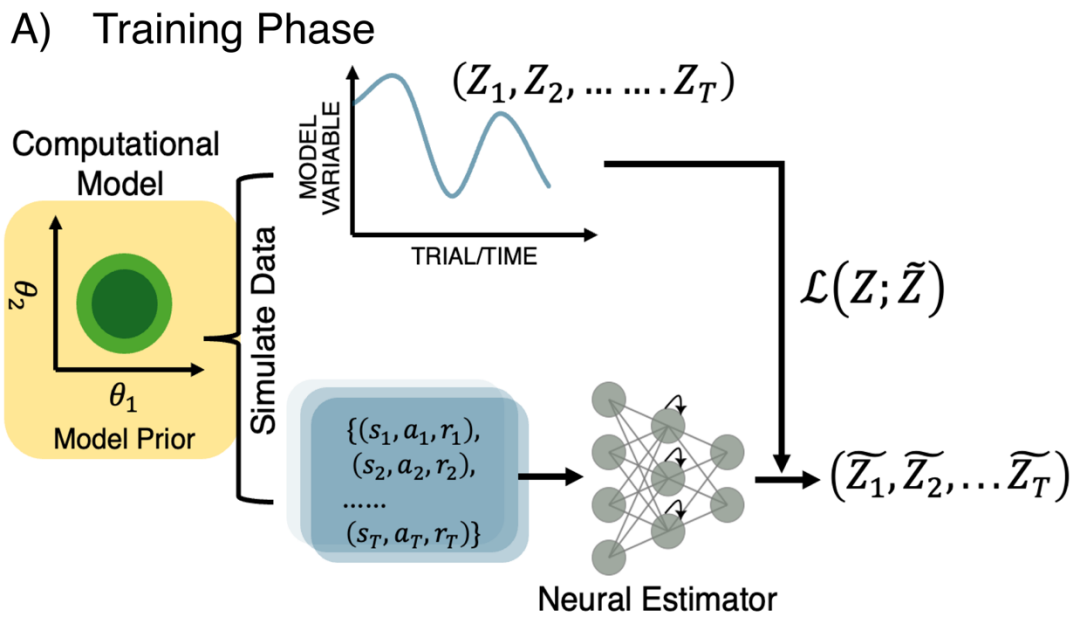
Approximate likelihood with…

- Complex mathematical formula (Ashwood et.al. 2021)
- Sampling method e.g., Sequential Monte Carlo/Particle Filtering. (Findling et.al, 2021)

Give up and develop a model with tractable likelihood

Proposed method

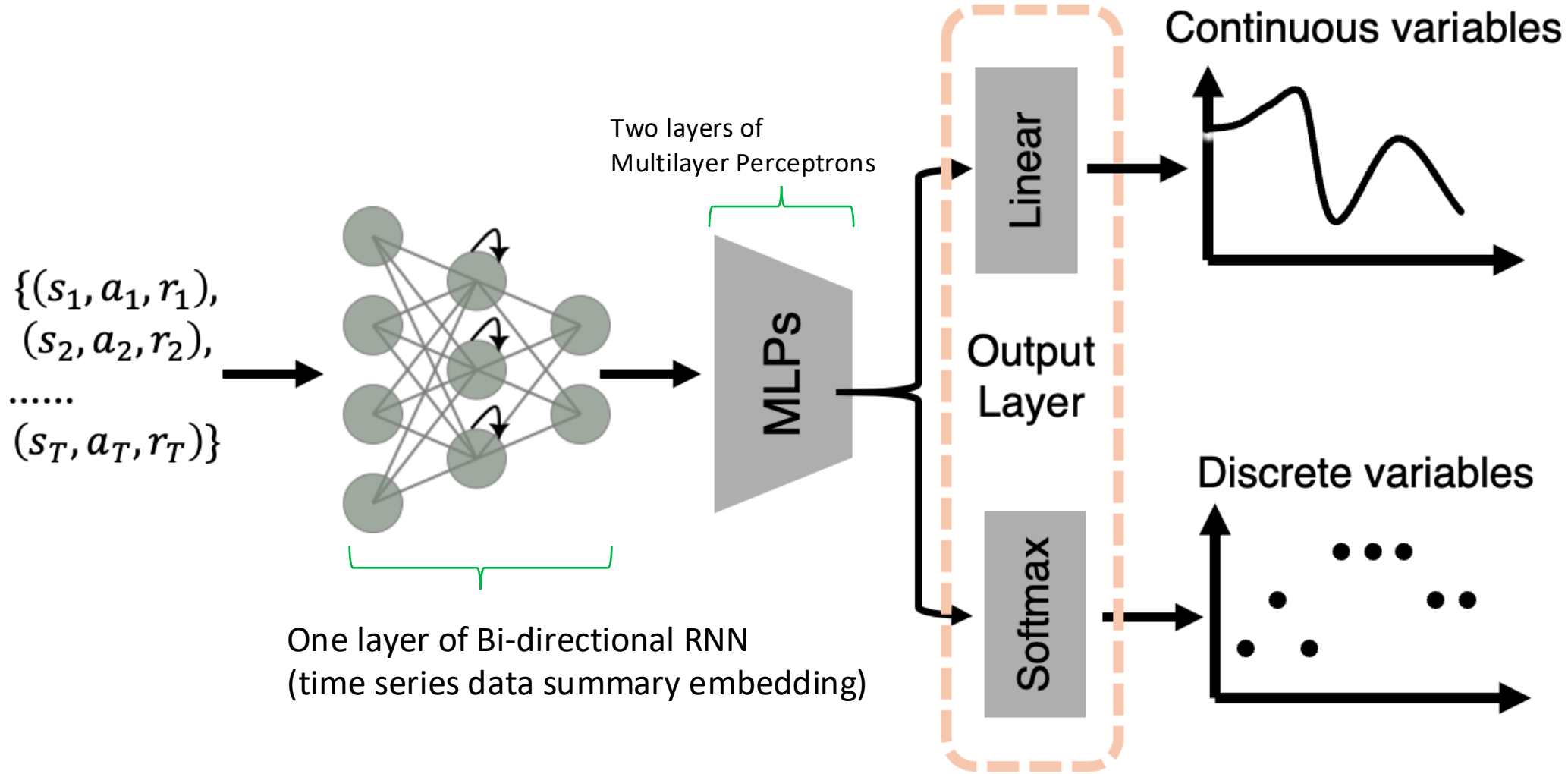
LaseNet: Latent Sequence identification neural Networks



Target

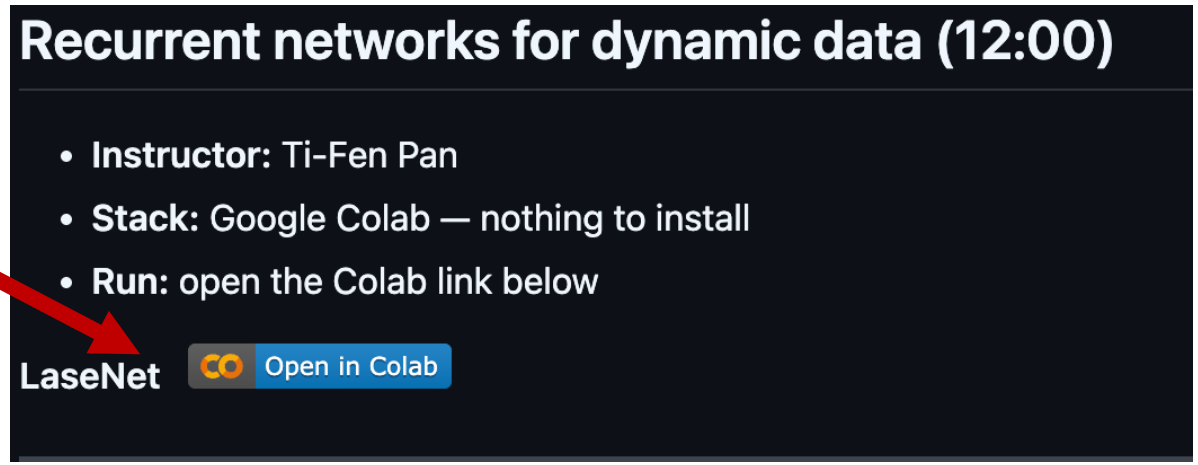
Joint Density: $P(z \mid \theta, Y)\ P(\theta \mid Y)$

Architecture - LaseNet



Your Turn!

1. Go to the folder “day 3 tools for SBI” on Github
2. In README, open the link under



3. **Go through the notebook from S0-S2 (before S3).**
4. **Discuss with 1 – 2 of your neighbors: what to fill in the blank at S3 (TODO)**

Acknowledgement

Jing-Jing Li

Bill Thompson

Anne GE Collins

Supported by the National Institutes of Health (NIH R21MH132974).

Slide design credits to Felo

Behavior Research Methods (2025) 57:272
https://doi.org/10.3758/s13428-025-02794-0

ORIGINAL MANUSCRIPT

Latent variable sequence identification for cognitive models with neural network estimators

Ti-Fen Pan[1] · Jing-Jing Li[2] · Bill Thompson[1] · Anne GE Collins[1,2]

Code available: https://github.com/ti55987/lasenet

Evaluating Performance Against Likelihood-Dependent Methods

Training time (mean ± std. dev. in minutes) of LaseNet on two likelihood-intractable models that predict both continuous and discrete latent variables. Each LaseNet was trained with a learning rate of 3e − 4 and a batch size of 256 on the L4 GPU provided by Google Colab.

Models	**HRL**				**Weber-imprecision**			
# of Trials	**720**				**300**			
# of Samples	**600**	**1000**	**3000**	**6000**	**600**	**1000**	**3000**	**6000**
Mean	4.25	6.28	17.36	28.44	2.72	3.51	7.84	12.02
Std. Dev	0.015	0.011	0.017	2.2	0.21	0.3	0.82	0.97

What can LaseNet do now?

Yes

- Identify latent variables within **likelihood-intractable models**.
- **Flexible** architecture that can handle both continuous and discrete latent spaces.
- **Generalizable** to a wide range of cognitive models (i.e., RL, Bayesian generative models, HMM)

No

- **Uncertainty quantification:** different loss functions e.g**.,** Kullback-Leibler (KL) Divergence Loss …etc
- Infer underlying **model parameters**: using LaseNet in conjunction with other methods.